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Causal Effects Without Randomization

“Perhaps consciousness arises when the brain’s simulation of the world becomes so complex that it must include a model of itself”

Richard Dawkins

Randomization is not always possible in practice. In the absence of randomization, determining causal effects becomes more complicated, since it requires more nuanced reasoning about counterfactuals and causality. In this chapter we will explore various statistical techniques for determining causal effects when randomization is not possible; these principles will be referenced in future chapters on Reinforcement Learning wherein we draw connections back to Causal Inference. Some indicative signposting is included for the Reinforcement Learning practitioner who may be already familiar with various concepts from optimal control theory. In short, models used for Causal Inference without randomization share similar purpose and structure as methods for offline Reinforcement Learning, and Reinforcement Learning where traditional assumptions are violated.

3.1 Confounding Effects

Recall Table 2.1, which consisted of four subjects (Claudius, Julius, Augustus and Octavian), where two are assigned to each treatment group. We previously computed an ATE by filling in the missing potential outcomes using groupwise means, noting that in order to do so, the treatment groups must be comparable, or *exchangeable*. However, the exchangeability criterion does not always hold in practice. Consider, for example, the case where drug 2 does not work for current smokers, but on former smokers it works better than drug 1. Suppose Claudius and Augustus are non-smokers, whereas Julius and Octavian are current smokers: Instead of being randomly assigned to treatments, smokers are given drug 1 while non-smokers are given drug 2. Then, the counterfactual estimates of Julius’ and Octavian’s outcomes had they been given drug 2 instead of drug 1 are no longer accurate.

The problem in the former smoker/current smoker scenario is that something has caused smokers to receive treatment 1 and non-smokers to receive treatment 2; that is, we have another variable X_m , which is entirely unobserved, and which drives both treatment assignment and causal effect. Since this missing variable X_m is an upstream causal influence on both the policy and the outcome, then Equation 2.3 does not hold. Figure 3.1 highlights this concept; here X_m would be an indicator function of smoking status, taking the value 1 if subject j is a smoker and 0 otherwise. These upstream variables are known as *con-*

founding effects: The issue of confounding is likely very familiar to both the practitioner of Reinforcement Learning as well as the Causal Inference expert. In Causal Inference Figure 3.1 is known as a *causal diagram*.

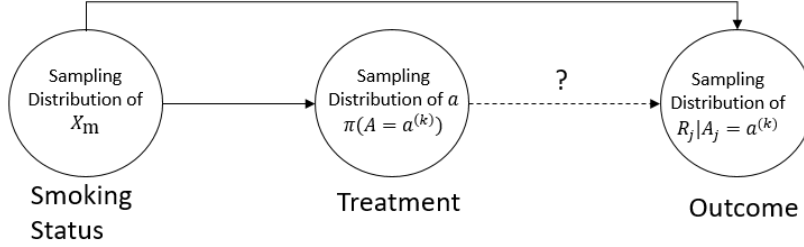


FIGURE 3.1: Causal diagram illustrating a confounded treatment effect. Here X_m is the unobserved confounder: In our example X_m corresponds to smoking status.

In the classic causal diagram, each node represents a variable, and an arrow $X \rightarrow Y$ encodes a direct causal effect of X on Y . The absence of an arrow between two nodes is itself a modeling claim, namely that the two variables have no direct causal relationship in the model under consideration (though indirect effects mediated through other variables in the diagram are still possible). Direction also matters: $X \rightarrow Y$ is a different claim from $Y \rightarrow X$. Variables not directly observed, such as the unmeasured confounder X_m in Figure 3.1, are sometimes drawn with a distinct visual treatment (a dashed border, a different shape, or shading) to flag that they are inferred rather than measured.

To the Reinforcement Learning Practitioner: Conditioning on smoking status to create a CATE is like conditioning on the observed state variables in a MDP. The CATE is the expected action-value given current state-action pair.

Suppose, for a moment, that the confounder was observed. Then, we might simply create a separate ATE for smokers and non-smokers; this version of the ATE is called a *conditional* ATE or CATE given smoking status.

A formal definition of the CATE is given in Definition 3.1: The CATE is typically estimated by computing groupwise means for each stratum of X .

Definition 3.1 (Conditional Average Treatment Effect (CATE)) *The Conditional Average Treatment Effect is the expected treatment effect within a specific subgroup of patients defined by their characteristics $X = x$. Empirically, one might create an estimate of the CATE by stratifying the data based on levels of X before averaging¹*

$$CATE(x) = \mathbb{E}[R|A = a^{(2)}, X = x] - \mathbb{E}[R|A = a^{(1)}, X = x] \quad (3.1)$$

¹Of course, stratification requires that the covariate X is categorical. We will address the case of continuous covariates later in this chapter.

Having defined the CATE, the data scientist might then begin to wonder, *under what conditions will the empirical CATE produce accurate estimates?* The groupwise mean approach to CATE estimation requires that the conditional expectations $\mathbb{E}[R \mid A = a, X = x]$, equal the causal estimands $\mathbb{E}[(R|a) \mid X = x]$. To ensure that this is the case, we need an additional assumption.

Recall from Chapter 2 the definition of *exchangeability* in the context of randomized experiments. In observational studies, treatment assignment is typically not random, but may be “as good as random” once we condition on a sufficient set of observed covariates (X , in our notation above). This generalized version of exchangeability is called *conditional exchangeability*, also known as *ignorability* or the *no unmeasured confounders* assumption. A definition is given in Definition 3.2.

Definition 3.2 (Conditional Exchangeability / Ignorability) *Treatment assignment A is conditionally exchangeable (or ignorable) given covariates X if the potential outcomes are independent of treatment assignment, conditional on X (Equation 3.2).*

$$\{R|a : a \in \mathcal{A}\} \perp A \mid X \quad (3.2)$$

In other words, conditional exchangeability means that within any stratum defined by $X = x$, the subjects who receive treatment $a^{(1)}$ are exchangeable with those who receive $a^{(2)}$ with respect to their potential outcomes. When it holds, observed differences in outcomes between treatment groups within strata of X can be attributed to the causal effect of treatment. When it fails (for instance, due to an unmeasured confounder X_m as in Figure 3.1), then the observed associations may not reflect true causal effects. The subsequent methods in this chapter, too, require conditional exchangeability to produce valid causal estimates.

A simple example of conditional CATE estimation for the smoking status confounder scenario follows. Suppose we observe eight subjects (four current smokers, four former smokers), each prescribed one of two drugs. Their observed outcomes are summarized in Table 3.1.

TABLE 3.1: Hypothetical observed outcomes for eight subjects.

Subject	Smoking Status	Treatment	Outcome
Claudius	Former	Drug 1	80
Julius	Current	Drug 1	76
Augustus	Former	Drug 2	100
Octavian	Current	Drug 2	55
Zeus	Former	Drug 1	81
Hera	Current	Drug 1	74
Aphrodite	Former	Drug 2	101
Artemis	Current	Drug 2	54

Computing groupwise means yields a naive ATE estimate that is essentially zero (Drug 1: 77.8 vs. Drug 2: 77.5). Conditioning on smoking status, however, reveals that there is a degree of *conditional* heterogeneity present in the data: for current smokers, Drug 1 actually outperforms Drug 2 (75.0 vs. 54.5), while for former smokers Drug 2 substantially outperforms Drug 1 (100.5 vs. 80.5). The R code that produces these estimates is available in the supplementary script `cate_example_simple.R`.

We can also combine the CATEs to create an unbiased ATE; if there are equal numbers of smokers and non-smokers, then the ATE is simply the average of the two CATEs. However,

if for example we have 60 smokers and 40 non-smokers, then the ATE would be $ATE = 0.6 \times \text{CATE of smokers} + 0.4 \times \text{CATE of non-smokers}$.

Readers planning to compute CATEs in practice may wish to use the `marginalEffects` R package [4], which provides a unified interface for predictions, contrasts, and marginal effects across a wide range of model classes. The manual subgroup-wise contrasts shown above can be replaced with a single call to `comparisons(model, variables = "A", by = "Smoking_Status")`, which also returns standard errors and confidence intervals out of the box.

To the RL practitioner: The ATE is similar to the value function. The difference is that the value function marginalizes over actions, while the ATE marginalizes over states.

The R function `lm()` allows us to flexibly control for covariates that may confound results; `lm()` performs ML estimation for linear Gaussian models. `lm()` is widely used across statistical applications, along with its extended versions `glm()` and `lmer()` which we will return to later. We will start with a very simple application of `lm()`, which performs a two-sample test for difference in means. For Python users, `lm()` is the R equivalent of `statsmodels.OLS`.

An R example follows, using a simple simulated dataset: Here, treatment with $a^{(2)}$ generally results in larger outcomes than $a^{(1)}$, but the advantage of $a^{(2)}$ over $a^{(1)}$ is larger in the non-smoker group than in the smoker group.

Controlling for Confounders

```
# lm_example.R

asthma_data <- data.frame(
  treatment = rep(c("a1", "a1", "a2", "a2"), 2),
  outcome = c(c(20, 25, 50, 45), c(20, 25, 50+10, 45+10)+5),
  smoking_status = c(rep("S", 4), rep("NS", 4))
)
# Model with interaction term
model_interaction <- lm(outcome ~ treatment * smoking_status,
                        data = asthma_data)

# This estimates separate treatment effects for each group
# Equivalent to learning different Q-functions for different contexts
```

The above code fits the model in Equation 3.3. Here, $\mu = \beta_0 + \beta_1 x_{trt,j} + \beta_2 x_{smok,j} + \beta_{12} x_{trt,j} \times x_{smok,j}$, where $x_{trt,j}$ takes the value 1 if the j^{th} subject received treatment 2, and 0 otherwise: $x_{trt,j} = \mathbb{1}(A_j = a^{(2)})$. This matches the coding `lm()` adopts by default, which treats the first factor level — here $a^{(1)}$ — as the reference. Similarly, $x_{smok,j}$ takes the value 1 if the j^{th} patient is a smoker, and 0 otherwise. r_j is the j^{th} patient's outcome, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The function `lm()` uses Maximum Likelihood estimation to produce estimates $(\hat{\beta}, \hat{\sigma}^2)$, where $\beta = (\beta_0, \beta_1, \beta_2, \beta_{12})$.

$$r_j = \beta_0 + \beta_1 x_{trt,j} + \beta_2 x_{smok,j} + \beta_{12} x_{trt,j} \times x_{smok,j} + \varepsilon_j \quad (3.3)$$

The non-smokers who received treatment $a^{(1)}$ serve as our reference group: The estimate $\hat{\beta}_0$ represents the average outcome among this population, which is why the fitted intercept

above is 27.500, the mean outcome among non-smokers on treatment 1. β_1 represents the *change* in expected outcome one might expect to see if a non-smoker was moved from treatment $a^{(1)}$ to $a^{(2)}$. β_2 represents how our predicted outcome would change for individual j if we found out that s/he was actually a smoker, having previously thought s/he was a non-smoker. Finally, the interaction term β_{12} , fit by `treatment:smoking_status`, allows the treatment effect to vary by smoking status; β_{12} represents the amount by which the treatment effect differs between smokers and non-smokers. Table 3.2 shows the expected outcomes for each combination of treatment and smoking status combination.

TABLE 3.2: Expected outcomes for each treatment-smoking status combination

	Non-smoker	Smoker
Treatment 1 ($a^{(1)}$)	β_0	$\beta_0 + \beta_2$
Treatment 2 ($a^{(2)}$)	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_{12}$

To get the CATE for each smoking status group, one would subtract the first row of Table 3.2 from the second row; for example, the CATE in the non-smoking group would be $(\beta_0 + \beta_1) - (\beta_0) = \beta_1$, and the CATE in the smoking group would be $(\beta_0 + \beta_1 + \beta_2 + \beta_{12}) - (\beta_0 + \beta_2) = \beta_1 + \beta_{12}$. Substituting the fitted values above, the non-smoker CATE is 35.0 and the smoker CATE is $35.0 - 10.0 = 25.0$. When interpreting the output of any linear modelling function, it is important to be aware of whether the function produces the raw parameter estimates $(\beta_0, \beta_1, \dots)$ or the compound estimates in Table 3.2. `lm()` returns the raw coefficients:

```
>summary(model_interaction)
```

Call:

```
lm(formula = outcome ~ treatment * smoking_status, data = asthma_data)
```

Residuals:

```
 1    2    3    4    5    6    7    8
-2.5  2.5  2.5 -2.5 -2.5  2.5  2.5 -2.5
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      27.500      2.500   11.000 0.000388 ***
treatmenta2       35.000      3.536    9.899 0.000584 ***
smoking_statusS   -5.000      3.536   -1.414 0.230200
treatmenta2:smoking_statusS -10.000      5.000   -2.000 0.116117
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 3.536 on 4 degrees of freedom
```

```
Multiple R-squared:  0.9762, Adjusted R-squared:  0.9583
```

```
F-statistic: 54.67 on 3 and 4 DF,  p-value: 0.001054
```

The function `lm()` assumes that the error terms on the specified model are independent and Gaussian; the set $\{\varepsilon_j\}$ must be an i.i.d. normal random sample. The function `glm()` relaxes the normality assumption, allowing error terms to have other distributions. For example, if the outcome is binary (say, response/no response), then the Gaussian assumption on the ε_j components is inappropriate; one should use the binomial distribution instead. Similarly, the function `lmer()` from the package `lme4` allows us to relax the independence

assumption somewhat; with `lmer()`, we can fit models where some covariates are random variables. We will discuss this type of approach in Chapter 6.

3.2 Missing Data

Let us return to the confounding variable scenario depicted in Figure 3.1. What if the subjects do not wish to disclose their smoking status? Then, the confounding variable is unobserved, similarly as in the potential outcomes table where we are “missing” the counterfactual observation where Claudius took drug 1 instead of drug 2. The confounding variable problem, like the potential outcomes problem, is at its heart an issue of missing data (see [110] for a thorough discussion). If it is not possible to observe the latent confounder, then we must consider what effect the missing data has on our model estimation methods.

Statisticians typically consider three types of missing data: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). This third type of missing data is also called *nonignorable* missing data; the term *ignorable* is used for data which is MCAR or MAR (see [98]).

Suppose that our censored dataset looked like Table 3.3, where the shaded cells are unobserved. Here, we are only missing three observations; two smokers and one non-smoker. Moreover, the values which have censored smoking status do not appear to be substantially different from the values which have smoking status observed. We say here that the data are *Missing Completely at Random*; the formal definition is given in Definition 3.3.

TABLE 3.3: Missing Completely at Random (MCAR): Shaded cells represent unobserved smoking status. Missingness is random with no relationship to treatment or actual smoking status.

Subject	Treatment	Smoking Status	Outcome r
Julius	$a^{(1)}$	Smoker	20
Crassus	$a^{(1)}$	Non-smoker	25
Sulla	$a^{(2)}$	Smoker	50
Augustus	$a^{(2)}$	Non-smoker	45
Brutus	$a^{(1)}$	Non-smoker	22
Cicero	$a^{(2)}$	Smoker	48
Cato	$a^{(1)}$	Smoker	24
Pompey	$a^{(2)}$	Non-smoker	52

Definition 3.3 (Missing Completely at Random (MCAR)) Let M be the missing data indicator matrix, where $M_{jk} = 1$ if the k^{th} entry for subject j is observed and 0 otherwise. In the potential outcomes setting of Chapter 2 the incomplete entries are the counterfactual outcomes, so $M_{jk} = 1$ exactly when $j \in J_k$ and $R_j|a^{(k)}$ is observed; in the confounding setting of this section the incomplete entries are covariate values such as smoking status. Data are Missing Completely at Random if missingness is unrelated to the data, observed or otherwise:

$$P(M \mid R, X) = P(M)$$

We can simulate MCAR data in R by deleting observations randomly. In the following simulation, we create a simple dataset consisting of three columns: Subject ID, treatment status, and smoking status.

Simulating Asthma Data with MCAR Missingness

```
# missing_types.R
set.seed(1998)
n <- 1000

df_oracle <- data.frame(
  Subject = 1:n,
  Treatment = sample(c("a1", "a2"), n, replace = TRUE),
  Smoking_Status = sample(c("Smoker", "Non-smoker"), n,
    replace = TRUE, prob = c(0.4, 0.6))
)

df_oracle$True_Outcome <- with(df_oracle,
  ifelse(Treatment == "a1",
    ifelse(Smoking_Status == "Smoker", 20, 22),
    ifelse(Smoking_Status == "Smoker", 25, 55))
) + rnorm(n, 0, 3)

Next, we delete 30% of the smoking status observations at random.

# MCAR
df_mcar <- df_oracle
missing_mcar <- sample(c(TRUE, FALSE), n,
  replace = TRUE, prob = c(0.3, 0.7))
df_mcar$Smoking_Status[missing_mcar] <- NA
```

If the data are MCAR, then our CATE estimates remain valid. We can see this in our simulation as follows:

CATE Estimation With MCAR Missingness

```
# missing_types.R

compute_cate <- function(data) {
  data_complete <- data[!is.na(data$Smoking_Status), ]

  ate_smokers <- mean(data_complete$True_Outcome[
    data_complete$Treatment == "a2" &
    data_complete$Smoking_Status == "Smoker"]) -
    mean(data_complete$True_Outcome[
      data_complete$Treatment == "a1" &
      data_complete$Smoking_Status == "Smoker"])

  ate_nonsmokers <- mean(data_complete$True_Outcome[
    data_complete$Treatment == "a2" &
    data_complete$Smoking_Status == "Non-smoker"]) -
    mean(data_complete$True_Outcome[
      data_complete$Treatment == "a1" &
      data_complete$Smoking_Status == "Non-smoker"])

  return(c(smokers = ate_smokers, nonsmokers = ate_nonsmokers))
}

true_cate <- compute_cate(df_oracle)

cate_mcar <- compute_cate(df_mcar)
result_mat = matrix(c(true_cate, cate_mcar), nrow=2, byrow=T,
  dimnames=list(c("True", "MCAR"), names(true_cate)))
result_mat
```

```
> result_mat
      smokers nonsmokers
True 4.538346   33.03472
MCAR 4.637825   33.12411
```

The MCAR estimates are very close to the true values based on the oracle data.

What if the dataset looks like Table 3.4? Here, most of the missingness in the smoking status column occurs among patients prescribed Treatment 2. However, the missingness within Treatment 2 affects smokers and non-smokers equally. In this situation, we say that the data are *Missing at Random*.

The formal definition of MAR data (Definition 3.4) requires some additional notation. Each column of the input covariate matrix can be partitioned into observed and missing parts: $X = (X_o, X_m)$, where X_o is the observed part of X and X_m is the missing part of X . Definition 3.4 means that the missingness can be explained by other observed variables in the dataset.

Definition 3.4 (Missing at Random (MAR)) *Data are Missing at Random if the probability of missingness depends only on the observed data and not on the missing data,*

TABLE 3.4: Missing at Random (MAR): Missingness concentrated in Treatment 2. Notice that within Treatment 2, both smokers and non-smokers are equally likely to be unobserved.

Subject	Treatment	Smoking Status	Outcome r
Julius	$a^{(1)}$	Smoker	20
Crassus	$a^{(1)}$	Non-smoker	25
Sulla	$a^{(2)}$	Smoker	50
Augustus	$a^{(2)}$	Non-smoker	45
Brutus	$a^{(1)}$	Non-smoker	22
Cicero	$a^{(2)}$	Smoker	48
Cato	$a^{(1)}$	Smoker	24
Pompey	$a^{(2)}$	Non-smoker	52

after controlling for the observed part of R as well as all fully observed covariates, as in Equation 3.4, where M is the missingness indicator matrix from Definition 3.3.

$$P(M \mid R_o, R_m, X_o, X_m) = P(M \mid R_o, X_o) \quad (3.4)$$

Even if data are censored by a MAR mechanism, the CATE estimates based on classical statistical models remain valid. This result generalizes to all models fit by maximum likelihood estimation (Theorem 3.1); here, we can embrace Johann Strauss II's idea expressed in *Die Fledermaus* that *Glücklich ist, wer vergisst, was doch nicht zu ändern ist* (that is, "Happy is he who forgets what cannot be changed") [162]. A numerical comparison of the MCAR and MAR estimates against the oracle truth is given in Table 3.6 later in this section.

Theorem 3.1 *If the missing data are ignorable, and if the model specification is correct, then the Maximum Likelihood Estimates computed from the observed data (R_o, X_o) alone are consistent for the parameters of the full-data model [98, 137].*

Various methods have been developed for efficiently fitting models to an incomplete dataset under the assumption that missingness is MAR. One popular method is the Expectation-Maximization (EM) algorithm [41], which iterates between an expectation step and a maximization step. Using the EM algorithm results in a more efficient analysis than discarding the missing rows like we have done previously. Algorithm 1 shows the EM algorithm; here Z_m represents the missing part of the data Z and Z_o is the observed part. As a reminder, the expectation operator $\mathbb{E}$ is defined as $\mathbb{E}[g(X)] = \int g(x)f(x) dx$ for a continuous random variable X with density f , and as $\mathbb{E}[g(X)] = \sum_x g(x)P(X = x)$ in the discrete case; the conditional expectation $\mathbb{E}_{Z_m|Z_o, \theta^{(j)}}[\cdot]$ in the E-step is the same operation applied to the conditional distribution of Z_m given Z_o and $\theta^{(j)}$. Two versions of θ also appear in the algorithm: $\theta^{(j)}$ denotes the parameter estimate at the current iteration j , while $\theta^{(j')}$ denotes a new potential candidate parameter value, which the algorithm considers in the M-step's search for the next iterate. Note that the E-step holds $\theta^{(j)}$ fixed and varies $\theta^{(j')}$, while the M-step picks the $\theta^{(j')}$ that maximizes the expected log-likelihood and promotes it to $\theta^{(j+1)}$.

To the Reinforcement Learning practitioner: The EM algorithm is similar in structure to the Value Iteration procedure; in Section 5.2 we will see that under certain specifications they can be identical.

Algorithm 1 Expectation-Maximization (EM) Algorithm

Require: Initial parameters $\theta^{(0)}$, observed data Z_o , convergence threshold ϵ , likelihood function f

- 1: $j \leftarrow 0$
- 2: **repeat**
- 3: // E-Step: Compute expected value of log likelihood
- 4: $\Phi(\theta^{(j')}|\theta^{(j)}) \leftarrow \mathbb{E}_{Z_m|Z_o, \theta^{(j)}}[\log f(Z_o, Z_m|\theta^{(j')})]$
- 5: // M-Step: Find parameters maximizing Φ
- 6: $\theta^{(j+1)} \leftarrow \arg \max_{\theta^{(j')}} \Phi(\theta^{(j')}|\theta^{(j)})$
- 7: $j \leftarrow j + 1$
- 8: **until** $\|\theta^{(j)} - \theta^{(j-1)}\| < \epsilon$
- 9: **return** $\theta^{(j)}$

Practitioners typically use the R package **Amelia** [67] to perform EM estimation. Most modern implementations of the EM algorithm, including **Amelia**, perform the E step using a method which models Z_m as a random variable, instead of a fixed quantity: Instead of estimating Z_m directly, we estimate the *distribution* of Z_m using Bayes' theorem, usually assuming that the missing part follows a Gaussian distribution or something similar. After model fitting, several samples are generated from Z_m 's distribution; these are combined in downstream analyses. We will return to more sophisticated Bayesian analyses in Chapter 6. An alternate approach to EM is the chained equation approach in the package **mice** [170].

A simple example of **Amelia** applied to the MAR dataset described earlier follows. The core call to **amelia()** produces m imputed datasets, each a single draw from the posterior of the missing values.

EM Imputation via Amelia

```
# em_amelia_example.R
amelia_imputed <- amelia(df_mar, m = 5,
                        noms = c("Treatment", "Smoking_Status"))
```

Standard practice is then to fit the analysis model separately on each of the m imputations and combine the parameter estimates using *Rubin's rule*, implemented in **mitools::MIcombine** [101]. The full pipeline (model fitting plus combination) is available in the supplementary script **em_amelia_example.R**. As indicated in the note to RL practitioners, we will see in Section 5.2 that the EM algorithm is connected to a foundational concept in Reinforcement Learning called *Value Iteration*.

What if the data are not MAR? We begin to encounter some difficulties when the missingness pattern resembles Table 3.5. Here, the patients with the worst outcomes are the least likely to report their smoking status (notice how the shaded cells belong to the subjects with the lowest values of r (20, 22, and 24)). Here, the probability of missingness depends on the outcome itself, even after conditioning on treatment and smoking status, and hence the complete cases are no longer a random subsample of any treatment-by-smoking-status group. As a result, averages computed from the observed samples only are systematically distorted. In this case, the data are referred to as Missing, Not at Random, or *non-ignorable*.

Definition 3.5 (Missing Not at Random (MNAR)) *Data are Missing Not at Random if the probability of missingness depends on the unobserved entries R_m or X_m even after conditioning on all observed outcomes and covariates:*

TABLE 3.5: Missing Not at Random (MNAR): Patients with poor outcomes are more likely to decline reporting their smoking status. Notice that the shaded (unobserved) cells belong to the subjects with the lowest outcomes.

Subject	Treatment	Smoking Status	Outcome r
Julius	$a^{(1)}$	Smoker	20
Crassus	$a^{(1)}$	Non-smoker	25
Sulla	$a^{(2)}$	Smoker	50
Augustus	$a^{(2)}$	Non-smoker	45
Brutus	$a^{(1)}$	Non-smoker	22
Cicero	$a^{(2)}$	Smoker	48
Cato	$a^{(1)}$	Smoker	24
Pompey	$a^{(2)}$	Non-smoker	52

$$P(M \mid R_o, R_m, X_o, X_m) \neq P(M \mid R_o, X_o)$$

In this case, the missingness mechanism depends on the missing values themselves, even after accounting for all observed data.

Under MNAR, the CATE estimates based on classical statistical models are no longer guaranteed to be valid; Theorem 3.1 does not apply because the missingness is no longer ignorable. The numerical comparison promised earlier is given in Table 3.6: the MCAR and MAR estimates remain unbiased, while the MNAR estimates depart systematically from the truth. Note that the MAR estimate wanders farther from the oracle value than the MCAR estimate does; this reflects variance rather than bias. The MAR mechanism discards sixty percent of the treatment-2 arm, so its complete-case estimate is computed from far fewer observations and is simply noisier—averaged over repeated simulations, its bias is zero, exactly as Theorem 3.1 predicts. The MNAR deviation, by contrast, does not average away: because patients with poor outcomes are missing more often, the surviving rows in every treatment-by-smoking-status cell over-represent good outcomes, and the estimates are pulled downward in every replication.

TABLE 3.6: CATE estimates under different missingness mechanisms, compared to the oracle truth (sample size $n = 1000$, seed 1998), as produced by the script `missing_types.R`. MCAR and MAR estimates are unbiased, as predicted by Theorem 3.1; the larger MAR deviation is sampling variance from discarding most of the treatment-2 arm, not bias. The MNAR estimates are systematically biased downward, because missingness depends on the outcome itself.

	Smokers CATE	Non-smokers CATE
True (oracle)	4.54	33.03
MCAR estimate	4.64	33.12
MAR estimate	5.03	33.66
MNAR estimate	3.68	32.00

The confounding variable scenario is a problem of nonignorable missingness. In the presence of an unobserved confounder, R_m depends on X_m . A downstream effect of this is that the treatment assignment probability is related to the outcome: $\pi(A = a^{(1)} | j = \textit{Julius}) \neq$

$\pi(A = a^{(1)} | j = \textit{Claudius})$, and thus Theorem 3.1 does not hold: Going back to the previous example, the unmeasured variable corresponding to smoking status was nonignorable, and thus we would be unable to correctly estimate the ATE based on Table 2.1. If, however, we recorded each patient’s smoking status and conditioned on it by including the “smoking status” variable in the model as a covariate, then maximum likelihood estimates of the ATE conditional on these values would be correct. It is important to note, however, that confounding effects are important even if we do have the ability to randomize: In practice, when an individual comes to us for treatment, we want to know what treatment works for that specific individual, even if this treatment is not optimal on average.

Exchangeability can also be framed in terms of non-ignorable missingness. Formally, treatment groups are exchangeable if the unobserved potential outcomes are ignorably missing (see [53]).

Recall from Chapter 2 the Monty Hall-type treatment selection problem. Here, when the game show host opens a door, the choice of *which* door to open depends on where the car actually is. From the perspective of the game show contestant, this is nonignorable missingness: The revelation mechanism depends on the unobserved values behind the doors. In the clinical version, discovering a contraindication for drug c is informative because the contraindication’s existence is related to the drug’s true quality (its side effect profile is part of what determines μ_c).

3.3 Inverse Probability of Treatment Weighting

Suppose we have data where treatment assignment *is* influenced by subject characteristics. All is not lost in this scenario; we can still estimate population-level treatment effects. The trick is to create a pseudo-population where treatment appears to be randomly assigned; this can be accomplished using a technique called Inverse Probability of Treatment Weighting (IPTW). By weighting observations based on their probability of treatment, IPTW effectively breaks the link between confounding variables and treatment assignment, which facilitates accurate estimation of causal effects from non-randomized data.

Let us return to the asthma example, where smoking status confounds the treatment effects. Previously, we simply stratified the data and computed a CATE, then combined these CATEs based on the population weights to create an ATE. However, this stratification is not always possible as the type and number of confounders increases. Consider, for example, the case where treatment is confounded by age, where the older you are the more likely you are to be treated. Stratification would require dichotomizing the age variable into categories, which reduces statistical power and can obfuscate results [152].

We present a simple example of a confounded dataset, where older subjects are more likely to receive treatment (age confounds treatment).

Simulating Confounded Treatment and Outcome

```
# age_confounded_iptw.R
set.seed(1998)

# Simulate confounded treatment and outcome
n <- 200
age <- rnorm(n, mean = 50, sd = 15)

# Treatment assignment depends on age (older more likely treated)
p_treat <- plogis(-5 + 0.1 * age)
treatment_num <- rbinom(n, 1, p_treat)
treatment <- factor(treatment_num,
                    levels = c(0, 1),
                    labels = c("Control", "Treatment"))

# Outcome depends on both age and treatment (true effect = 10)
outcome <- 50 - 0.3 * age + 10 * treatment_num + rnorm(n, 0, 10)

data <- data.frame(id = 1:n, age = age,
                  treatment = treatment, outcome = outcome)
```

For binary treatments, subjects who received treatment ($A_j = 1$) are weighted by $\frac{1}{\pi(1|X_j)}$, while subjects who did not receive treatment ($A_j = 0$) are weighted by $\frac{1}{1-\pi(1|X_j)}$. The intuition behind this approach is straightforward: Subjects who had a low probability of receiving their actual treatment (based on their characteristics) are underrepresented in the observed sample and thus receive higher weights to compensate. The formal definition of an IPTW estimator for the Average Treatment Effect (ATE) is given in Equation 3.5, where X_j, R_j are the covariates and outcome, respectively, from the j^{th} subject.

$$\widehat{ATE}_{IPTW} = \frac{1}{n} \sum_{j=1}^n \frac{A_j R_j}{\pi(1|X_j)} - \frac{1}{n} \sum_{j=1}^n \frac{(1 - A_j) R_j}{1 - \pi(1|X_j)} \quad (3.5)$$

For a subject j with covariates X_j who received treatment A_j , the weight can be computed using Equation 3.6, where $\pi(A_j|X_j)$ is the probability of receiving treatment A_j given covariates X_j . This probability, also called the propensity score, can be estimated using logistic regression or other classification methods.

$$w_j = \frac{1}{\pi(A_j|X_j)} \quad (3.6)$$

The propensity scores and IPT weights can be computed using the following code. We use the function `glm` to fit the propensity score model: `glm` fits a *generalized linear model* to the data. The model is *generalized* because it allows flexible distributional assumptions on the error terms: Since treatment is binary, we use the binomial distribution instead of the Gaussian. Standard linear models can also be computed with `glm` using `family = gaussian()`. In the following code block, `ps_model` is the propensity score model, and `ps` contains the propensity scores.

Estimating Propensity Scores

```
# age_confounded_iptw_base.R
# Calculate propensity scores and IPT weights
model_ps <- glm(treatment ~ age, data = data, family = binomial())
ps_raw <- predict(model_ps, type = "response")

# Probability of receiving the actually-observed treatment
data$ps <- ifelse(data$treatment == "Treatment", ps_raw, 1 - ps_raw)

# Inverse probability weights
data$ipw <- 1 / data$ps
```

Next, we can compute the treatment effect estimates.

Treatment Effect Estimates: Naïve vs IPTW

```
# age_confounded_iptw.R
# Naive group means
naive_mean <- tapply(data$outcome, data$treatment, mean)

# IPTW-weighted group means: sum(outcome * ipw) / sum(ipw) per arm
weighted_mean <- with(data,
  tapply(outcome * ipw, treatment, sum) / tapply(ipw, treatment, sum)
)

# Differences (Treatment minus Control)
naive_diff <- diff(naive_mean)
weighted_diff <- diff(weighted_mean)

true_effect <- 10
cat("Naive estimate:", round(naive_diff, 2), "\n")
cat("IPTW estimate:", round(weighted_diff, 2), "\n")
cat("True effect:", true_effect, "\n")
```

Running the above code, we see that the IPT weighting strategy effectively unbias the treatment effect estimate.

```
Naive estimate: 3.99
IPTW estimate: 10.75
True effect: 10
```

One challenge with IPTW is that extreme weights can occur when treatment probabilities are very small, leading to high variance in the estimator. Suppose, for example, we extend our age-confounding scenario to include a subpopulation of very elderly patients (age > 80). Clinical guidelines strongly recommend treatment for this group, so physicians prescribe treatment to elderly patients with probability ≈ 0.95 . However, suppose one 82-year-old patient, call her Livia, declined treatment due to concerns about side effects.

Livia's IPTW weight would be $w_{\text{Livia}} = \frac{1}{1-0.95} = 20$. This single observation receives twenty times the weight of a typical patient. If Livia happens to have an unusually good or bad outcome for reasons unrelated to treatment (perhaps she has excellent diet and

exercise habits, or conversely has multiple comorbidities), her extreme weight will introduce substantial variance into our ATE estimate.

We can simulate this pathological scenario by modifying our previous age-confounding example to include strong age-treatment relationships as follows.

Simulating Extreme Weights Scenario

```
# extreme_weights_example.R

set.seed(1998)

# Include some very young and very old patients
age <- c(rnorm(100, mean = 50, sd = 15),
        rnorm(100, mean = 25, sd = 3),    # Very young
        rnorm(5, mean = 85, sd = 2))     # Very old

n <- length(age)

# Strong age-treatment relationship
p_treat <- plogis(-8 + 0.15 * age)

treatment <- rbinom(n, 1, p_treat)
outcome <- 50 - 0.3 * age + 10 * treatment + rnorm(n, 0, 10)

# Compute IPTW weights
ps_model <- glm(treatment ~ age, family = binomial())
ps <- predict(ps_model, type = "response")
weights <- ifelse(treatment == 1, 1 / ps, 1 / (1 - ps))

# Check for extreme weights
summary(weights)
max(weights)
```

```
> summary(weights)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 1.002  1.028   1.042   1.873   1.386  29.649
> max(weights)
[1] 29.64866
```

In this modified scenario, the maximum weight approaches 30, meaning that a single patient's outcome can dominate the weighted average. The resulting ATE estimate, while theoretically unbiased, has enormous variance and is practically unreliable.

To address this, stabilized weights are often used. The formula for these stabilized weights is given in Equation 3.7, where $\pi(A_j = a_j)$ is the marginal probability of receiving treatment a_j .

$$sw_j|a_j, x_j = \frac{\pi(A_j = a_j)}{\pi(A_j = a_j|X_j = x_j)} \quad (3.7)$$

IPTW relies on two critical assumptions: i) every subject or sample must have a non-zero probability of receiving each possible treatment, and ii) there must not be any unmeasured confounders. Assumption i) is known as *positivity*, which we formally define as follows.

Definition 3.6 (Positivity) For all values of covariates x that occur in the population (i.e., where $P(X = x) > 0$), positivity is satisfied if and only if every treatment level has a nonzero probability of being assigned. That is, we say that positivity is satisfied iff Equation 3.8 holds.

$$0 < \pi(A = a \mid X = x) < 1 \quad \text{for all } a \in \mathcal{A} \text{ and all } x \text{ with } P(X = x) > 0 \quad (3.8)$$

Positivity is also sometimes called the *overlap* or *common support* condition. If positivity is violated, as in the case where elderly patients are always treated and never serve as controls, then the inverse probability weights $1/\pi(A \mid X)$ can be infinite, leading to the extreme-weight pathology described above.

The Three Identifiability Assumptions for Causal Inference

Three assumptions are typically required to identify causal effects from observational data. These assumptions are:

1. **Conditional Exchangeability / Ignorability** (Def. 3.2): The potential outcomes are independent of treatment assignment, given observed covariates: $\{R|a\} \perp A \mid X$.
2. **Positivity** (Def. 3.6): Every subject has a nonzero probability of receiving each treatment: $0 < \pi(A = a \mid X = x) < 1$.
3. **Consistency** (Def. 2.4 in Chapter 2): If a subject receives treatment $A = a$, then the observed outcome equals the potential outcome under a : $A = a \implies R = R|a$.

These assumptions, along with SUTVA (Def. 2.3 from Chapter 2), support all of the causal estimation methods presented in this chapter. See also [62, 69].

Brazilian Jiu Jitsu (BJJ) is a grappling sport in which two practitioners attempt to control and submit each other on the ground; a “submission” is a hold, typically a joint lock or a strangle, that ends the match when the opponent taps out. We will use BJJ as a recurring example throughout this book, returning to it in multiple cases where the math calls for actions whose effects depend on circumstances the analyst cannot fully observe.

Suppose I wish to estimate what submissions are most effective for me. One of my favorite submissions is the triangle armbar, wherein you trap the opponent’s head and a single arm between your legs, and then hyperextend the trapped arm, resulting in exorbitant pressure on the elbow. Other arm attacks include Kimura, omoplata, and Americana. Another submission I like is the guillotine, which involves threading one’s arm around the opponent’s neck and strangling them. I am also working on improving my ankle locks, which - if applied successfully - result in the opponent’s ankle being bent the wrong way. Readers who are not BJJ practitioners can safely treat the submission names as opaque labels, though many informative instructionals exist on YouTube and elsewhere.

Whether or not I am tired might play a role in which submissions I attempt. For example, let’s say I find myself in bottom side control, which is a very bad position where my opponent is lying perpendicularly across me. Getting out of side control, particularly from under a bigger opponent, takes a massive amount of effort. So, if I am exhausted, I might attempt a last-resort buggy choke instead of trying to escape. The buggy choke has a very low probability of success, but if I have no energy to get out of bottom side control, then it might be my only option.

The attentive reader may notice that the structure of this example is the same as treatment selection in a non-randomized clinical trial. I have a small set of actions, with varying

baseline success rates, and a confounding variable (fatigue) that affects both which action I choose and how likely I am to succeed. The only difference is that the actions are martial arts moves and the decision-maker is one specific (fatigued) author.

To determine which of these attacks works the best for me, I might play a set of rounds and record the number of successful submissions of each type. Here, each submission corresponds to a “treatment”, and submission success is analogous to patient outcome in the clinical trials example. We wish to estimate the ATE for each submission. Concretely, the estimand for each submission is the success rate it would achieve if I attempted it in every round, averaged over how often I am fresh versus fatigued. This is the quantity a naive per-submission success rate will fail to recover, because I do not choose submissions independently of my fatigue. We can simulate this situation in R, and estimate the success probabilities of each submission, using a propensity score model with fatigue as a predictor.

First we define base success rates: I am most successful with triangles, and least successful at buggy chokes. We also fix the penalty that fatigue imposes on success: when I am exhausted, every submission is half as likely to work.

Simulating BJJ Submission Data

```
# bjj_submission_sw_base.R

set.seed(1998)

base_success <- c(AnkleLock = 0.7, Triangle = 0.9,
Kimura = 0.5, BuggyChoke = 0.05)

# Fatigue multiplies every submission's success probability by this
fatigue_penalty <- 0.5
```

Next we define the base observation rates. Usually I attempt triangles a lot (70 attempts here), followed by ankle locks and Kimuras (50 attempts), and buggy chokes rarely (1 attempt).

```
observations = c(rep("AnkleLock", 50), rep("Triangle",70),
rep("Kimura",50), rep("BuggyChoke",1))
```

If I am fatigued, I attempt buggy chokes more often (20 attempts). We then repeat this whole training pattern over many sessions, so that the rare fresh buggy-choke attempts accumulate enough data to estimate.

```

fatigued_observations = c(rep("AnkleLock", 40), rep("Triangle",50),
rep("Kimura",30), rep("BuggyChoke",20))

sessions <- 500

#Fatigue indicator
fatigue = rep(c(rep(0, length(observations)),
rep(1, length(fatigued_observations))), times = sessions)

submission <- factor(rep(c(observations, fatigued_observations),
times = sessions), levels=names(base_success))
n = length(submission)

```

Next we define the success probabilities and simulate the binary outcome (1 = success). Note that fatigue enters here as well as in the choice of submission: an exhausted attempt succeeds at half the rate of a fresh one. This second arrow is what makes fatigue a genuine confounder rather than a harmless driver of action selection. If fatigue only changed which submission I attempted, without changing whether it worked, there would be nothing for the weighting to correct.

```

success_prob <- base_success[as.character(submission)] *
ifelse(fatigue == 1, fatigue_penalty, 1)

outcome <- rbinom(n, size = 1, prob = success_prob)

```

Because we built the data generating process ourselves, we can write down the estimand exactly, and compare it against the naive per-submission success rate.

```

# Success rate each submission would achieve if always attempted,
# averaged over the marginal fatigue distribution
p_fatigued <- mean(fatigue)
true_marginal <- base_success *
((1 - p_fatigued) + p_fatigued * fatigue_penalty)

# Naive estimate: raw success rate per submission
naive_est <- tapply(outcome, submission, mean)

```

We estimate the propensity scores using a multinomial logistic model, because the action — the submission I choose — has four levels rather than two. Note that it is the number of levels of the *action* that dictates this choice; the propensity model describes treatment assignment, so the type of the outcome plays no part in selecting it. Had I been choosing between only two submissions, ordinary logistic regression via `glm()` would have sufficed. The multinomial distribution is not included in the options of `glm()`, so we use `multinom()` from the package `nnet` instead [175].

```

library(nnet)
ps_model <- multinom(submission ~ fatigue, trace = FALSE)
ps <- predict(ps_model, type = "probs") # n x 4 matrix

```

The naive estimate goes badly wrong for the buggy choke, reporting a success rate of 0.0250 against a true marginal rate of 0.0387 — an underestimate of roughly a third. The reason is the near-total class separation on fatigue status: buggy chokes are attempted almost exclusively in the fatigued state, so their raw success rate measures the exhausted version of the move rather than the move itself. The same mechanism biases the other three submissions in the opposite direction, though far more mildly. Kimuras and triangles are attempted disproportionately when I am fresh, so their raw rates (0.4084 and 0.7129) sit slightly above the truth (0.3875 and 0.6974). Ankle locks happen to be attempted at almost exactly the population fatigue rate, and their naive estimate is correspondingly almost unbiased.

In the BJJ example, the assumption of positivity means that no submission-fatigue state pair is impossible; I might still, for example, choose to do the buggy choke as a joke when rolling against a less-skilled player or a friend. Absence of unmeasured confounding demands that all variables that affect both treatment assignment and outcome are measured and included in the propensity score model. Implementing these on our BJJ dataset, we can see that the stabilized weights substantially correct the confounding.

Stabilized Weights for BJJ Submission Data

```
# bjj_submission_sw_base.R

# Propensity of the actually-chosen action
submission_int <- as.integer(submission)
pi_x <- ps[cbind(1:n, submission_int)]

# Marginal probability of each submission level
pi_marginal <- table(submission) / n
pi_a <- as.numeric(pi_marginal[as.character(submission)])

# Stabilized weights, then weighted success rate per submission
sw <- pi_a / pi_x
weighted_est <- tapply(outcome * sw, submission, sum) /
  tapply(sw, submission, sum)

result <- rbind(
  'True marginal' = true_marginal,
  Naive          = naive_est[names(base_success)],
  Stabilized     = weighted_est[names(base_success)]
)
print(round(result, 4))
```

	AnkleLock	Triangle	Kimura	BuggyChoke
True marginal	0.5424	0.6974	0.3875	0.0387
Naive	0.5442	0.7129	0.4084	0.0250
Stabilized	0.5422	0.6979	0.3894	0.0350

The three well-supported submissions are recovered almost exactly: 0.5422, 0.6979 and 0.3894 against true values of 0.5424, 0.6974 and 0.3875. The buggy choke improves from 0.0250 to 0.0350, closing most of the gap to 0.0387 but not all of it. That residual is the near-violation of positivity making itself felt. Fresh buggy-choke attempts make up well under one percent of my non-fatigued rounds, so the weighting must scale each of them up

by a factor of roughly twelve, and the estimate inherits the sampling noise of that small, heavily up-weighted cell. The remaining gap here is within about one standard error of the truth; had I never once attempted a buggy choke while fresh, no amount of weighting could have recovered the estimate at all.

IPTW is a special case of importance sampling, which is a popular tool deployed in Reinforcement Learning applications to the same end. We will revisit this connection later.

3.4 Marginal Structural Models for Time-Varying Treatments

Adaptive designs for clinical trials have gained popularity in recent years. For example, instead of assigning treatment arms equiprobably as in the conventional RCT one might modify the treatment-assignment probabilities to favor the treatments which look like they are good performers. These types of trials differ from the conventional RCT in the order that we act as judge and juror: In the adaptive trial, we return to the juror position at least once after acting as judge, while in the RCT we do not, acting as judge and juror only one time each, with the juror preceding the judge. One popular application of adaptive clinical trials is in cancer research: A book by Kosorok and Moodie [88] describes these various applications in detail.

While I have led with a particular example for clarity, the general “adaptive trial” is a broader category. The trial which modifies the treatment-assignment probabilities as suggested above (a process properly called *response-adaptive randomization*), is one subclass of adaptive trial among several. Other common flavors of adaptive design include early-stopping rules (where the trial terminates as soon as an interim analysis crosses a pre-specified threshold, an example of which will be introduced shortly), adaptive sample-size rules (where the planned enrollment is revised mid-trial based on the observed effect size or variance), dose-finding designs (where the dose level for each new subject depends on outcomes observed in earlier subjects, as in the 3+3 and continual reassessment method designs common in oncology phase I trials), and adaptive enrichment (where eligibility criteria are tightened mid-trial to focus on a subgroup identified as more responsive). In all of these types of adaptive trials, decisions are made based on preliminary data, rather than only at the end. The formal definition of an adaptive trial is given below.

Definition 3.7 (Adaptive Clinical Trial, Interim Analysis) *Any clinical trial which involves making decisions based on preliminary data before the study end is an Adaptive Clinical Trial. Analyses performed on these preliminary data are called interim analyses.*

A simple adaptive clinical trial design might involve performing an interim analysis on the data once we have seen responses from half of the data (Figure 3.2). If the test statistic computed from the incomplete dataset is larger than some pre-specified threshold, then we terminate the study early and assign all future subjects to the better treatment.

One famous example is the Multicenter Automatic Defibrillator Implantation Trial (MADIT), which was designed to test whether prophylactically implanting a cardioverter-defibrillator in high-risk patients — those who had survived a heart attack and had both reduced cardiac function and episodes of abnormal heart rhythm — reduced mortality relative to conventional therapy with antiarrhythmic drugs [112]. Stakes were extremely high, since the efficacy of each approach was measured by counting the number of deaths in each treatment arm. The study had originally assigned 95 patients to the defibrillator group, and 101 patients to the conventional therapy group. An interim analysis was performed at 27

months, wherein there were 39 deaths in the conventional therapy group and only 15 in the defibrillator group. Since the defibrillator treatment was clearly better, it became unethical to continue to enroll patients into the conventional therapy group, and so the study was terminated.

A slightly more sophisticated response-adaptive design for a clinical trial is called *play-the-winner*. In play-the-winner studies, subjects are assigned to treatments based on the previous subject's outcome. Consider the two-arm clinical trial in the case where outcomes are binary; we classify responses into “success” or “failure”. Play-the-winner treatment assignment works as follows:

1. Randomize the treatment arm for the first subject.
2. If the previous subject responded, then assign this subject to the same treatment arm as the previous subject. Otherwise, assign this subject to the other treatment.
3. Return to 2. Repeat for all future incoming subjects.

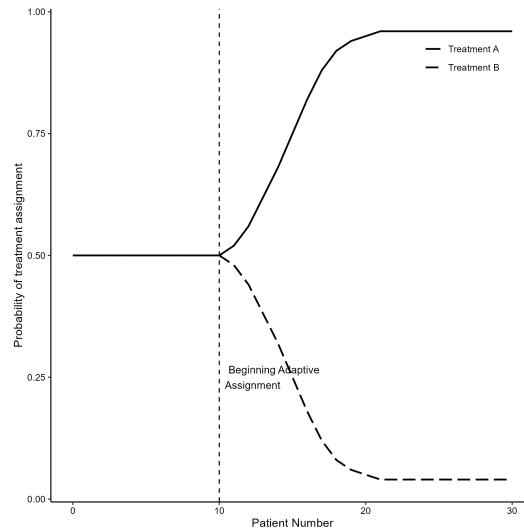


FIGURE 3.2: Treatment assignment probabilities should change with clinical outcome

Sequential analysis was founded by Wald [177] for industrial quality control during World War II; the modern group sequential designs used in clinical trials, such as the Pocock boundary [121], are direct descendants. Adaptive clinical trials, therefore, form one branch of a broader literature on adaptive experimentation; indeed, there are parallel methods in seemingly disparate fields such as econometrics and market research. A framework called the multi-armed bandit is also popular in all of these situations. We treat bandits directly in Chapter 4, and refer readers interested specifically in clinical-trial applications to [176].

The policy-choice experiment [81], created by economists, is one such analogous framework. In economics, where treatments are policy choices instead of chemotherapy regimens, it is plausible to maximize the eventual discovered policy at the expense of some early failures. The policy-choice experiment uses an allocation rule called “exploration sampling,” which is designed with this premise; the goal is to identify the best treatment efficiently, even if it means trying some bad policies early on. Exploration sampling is closely related to a method called Thompson sampling that we will encounter in the bandit chapters of Part II.

In market research, adaptive choice-based conjoint (ACBC) is a popular tool; these adaptive surveys allow earlier responses to determine later questions. Note, however, that the adaptation here occurs within-subject rather than between-subject, as in the adaptive clinical trial frameworks we have discussed earlier. That is, in ACBC, a respondent's later questions depend on their own earlier answers, not on outcomes from earlier respondents [168]. The clinical-trial analogue would be a phase I dose-finding trial where doses for each patient depend on that patient's earlier observed responses.

Indeed, some of the more sophisticated adaptive trials not only evolve across patient iterations, but timepoint iterations as well; sequential treatments that evolve through time pose additional statistical issues. Marginal structural models (MSMs) were designed to resolve these issues [133].

Time-dependent confounding is problematic because it violates the simplistic assumptions of standard regression methods.

Definition 3.8 (Time-dependent Confounder) *A time-dependent confounder X_m is a variable that satisfies the following conditions:*

- X_m is affected by previous treatments
- X_m influences subsequent treatment decisions
- X_m Predicts the outcome

Consider, again, the case of a physician treating an asthma patient over time. Here, the rate of asthma exacerbations serves as an outcome measure; the goal is to reduce the frequency of such exacerbations. However, the asthma exacerbation rate is also a key determinant of treatment decisions: When a patient experiences an increase in exacerbations, the doctor intensifies the therapy regimen in response (Figure 3.3). The result is a violation of the exchangeability assumption: In our table of potential outcomes for each patient, once again the missing items differ substantially from the non-missing items.

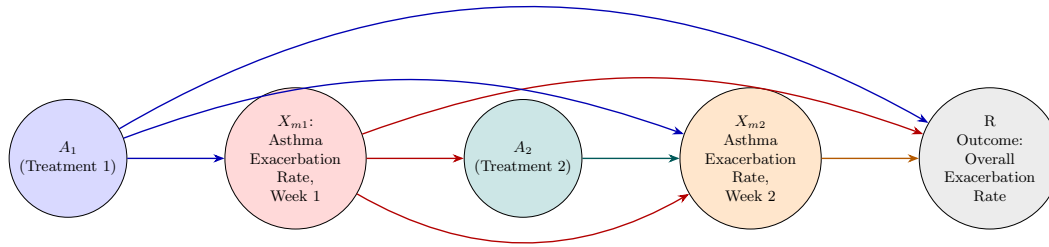


FIGURE 3.3: Causal diagram depicting a treatment effect with time-dependent confounding.

We can simulate this situation in R as follows.

Asthma Exacerbation Data with Time-Dependent Confounding

```
# msm_asthma.R
set.seed(8675309)

n <- 5000 # patients
TT <- 6   # monthly visits
```

We assume that the weekly asthma exacerbation count comes from a Normal-Poisson mixture, with the resulting count clipped at zero.

```
# Baseline weekly exacerbation count
baseline_exac <- pmax(rpois(n, lambda = 4) + rnorm(n, 0, 0.5), 0)

# Long-format panel: one row per (patient, month)
d <- data.frame(
  id      = rep(1:n, each = TT),
  t       = rep(0:(TT - 1), times = n),
  baseline_exac = rep(baseline_exac, each = TT)
)
```

The simulation loop below walks through each visit. At each step, the intensified-treatment indicator is drawn from a logistic model in the current exacerbation rate (doctors intensify when patients are doing poorly), and the next exacerbation count is drawn from a Normal model in the current treatment. After the loop, a one-step treatment lag is computed within each patient using `ave()`.

```

# Pre-allocate columns
d$exac_prev      <- NA_real_
d$intensified_tx <- NA_integer_
d$exac           <- NA_real_

# At t=0, prior exacerbation equals baseline
d$exac_prev[d$t == 0] <- d$baseline_exac[d$t == 0]

for (k in 0:(TT - 1)) {
  idx <- d$t == k

  d$intensified_tx[idx] <- rbinom(
    n      = sum(idx),
    size = 1,
    prob = plogis(-2 + 0.4 * (d$exac_prev[idx] - 3))
  )

  d$exac[idx] <- pmax(
    d$exac_prev[idx] + rnorm(sum(idx), 0.3, 1.2)
    - 2.0 * d$intensified_tx[idx],
    0
  )

  if (k < TT - 1) {
    d$exac_prev[d$t == k + 1] <- d$exac[d$t == k]
  }
}

# One-step treatment lag, by patient
d <- d[order(d$id, d$t), ]
d$tx_lag1 <- ave(d$intensified_tx, d$id,
  FUN = function(x) c(0, x[-length(x)]))

```

It would be disingenuous to apply classic statistical methods to this dataset. For example, if one were to adjust for time-varying covariates affected by previous treatment, one would be “controlling away” part of the treatment effect (a variable that is itself affected by earlier treatment is a mediator, and adjusting for it removes the part of the effect that travels through it; if the variable is additionally a collider, the adjustment can open a non-causal path as well [117]. We will return to both mediators and colliders in the section on Directed Acyclic Graphs). Conversely, failing to adjust for these time-varying covariates leaves us vulnerable to unmeasured confounding.

MSMs circumvent the confounding variable issue by creating a pseudo-population where treatment assignment is sequentially randomized at each time point: This pseudo-population is created by re-weighting each observation conditional on the treatment history and covariates at each timepoint. The MSM approach builds upon the IPTW method discussed in Section 3.3, applying it recursively through the time series.

The stabilized weights for the i^{th} subject at time t are calculated as $SW_i(t) = \prod_{t'=0}^t \frac{\pi(A_{it'}=a_{it'}|\bar{A}_{i(t'-1)})}{\pi(A_{it'}=a_{it'}|\bar{A}_{i(t'-1)},\tilde{X}_{it'})}$. Here, $a_{it'}$ represents the observed treatment for subject i at time t' , $\bar{A}_{i(t'-1)}$ represents the treatment history through time $t' - 1$, and $\tilde{X}_{it'}$ represents

the covariate history through time t' .

The numerator provides stabilization to reduce the variability of the weights, and the denominator accounts for the confounding by treatment history and time-varying covariates. Once these weights are calculated, one may estimate the parameters of a marginal structural model as in Equation 3.9, where $R|a$ represents the potential outcome under treatment regime $\bar{a}$, and X_{i0} represents baseline covariates.

$$E[R_{it}|A_{it} = a] = \beta_0 + \beta_1 \sum_{t'=0}^t a_{t'} + \beta_2 X_{i0} \quad (3.9)$$

In addition to the time-series correlation induced by treatment (dotted line in Figure 3.3), there is usually also a direct time-series correlation between each exacerbation rate and its predecessor. We can account for this by allowing the error terms on each observation r_{it} to correlate with one another across time, assuming that the correlation $\text{corr}(r_{it}, r_{i,t+1})$ between successive observations is equal to some constant β_ρ . This is called an *autoregressive* or AR(1) model (if we allowed some additional correlation between r_{it} and $r_{i,t+2}$, it would be an AR(2) model, and so on).

We fit four models to the asthma data. The first applies the MSM weights but treats the repeated observations within a patient as independent, so it corrects the time-dependent confounding without modelling the autocorrelation. We use the function `ipwpoint` from the package `ipw` to compute the propensity score weights, and `geeglm` to fit the weighted outcome model. Note that the binomial family here belongs to the *propensity* model, whose response is the binary treatment indicator; the outcome being modelled by `geeglm` is the exacerbation count, which is continuous, so that model is Gaussian. Note also that `ipwpoint` treats each row as an independent point exposure. This is adequate here because the weights are rebuilt from the one-step lag at every visit, but for a fully time-varying treatment the package's `ipwtm` function computes the cumulative product weights of the formula above directly.

```
library(ipw)
library(geepack)
wt <- ipwpoint(exposure = intensified_tx,
               family    = "binomial",
               link       = "logit",
               numerator  = ~ tx_lag1,
               denominator = ~ tx_lag1 + exac_prev,
               data       = d)
d$sw <- wt$ipw.weights

# Cumulative treatment exposure within each patient
d$cum_tx <- ave(d$intensified_tx, d$id, FUN = cumsum)

# MSM with independence correlation structure
msm_fit <- geeglm(exac ~ cum_tx + I(t/2) + baseline_exac,
                  id     = id,
                  weights = sw,
                  corstr  = "independence",
                  data    = d)
```

The second model reverses the emphasis: it imposes the AR(1) correlation structure but omits the MSM weights, so it models the within-patient correlation while leaving the

time-dependent confounding uncorrected. The function `geeglm` can also handle such perturbations to the variance structure.

```
# AR(1) correlation without weights (still confounded)
ar1_fit <- geeglm(exac ~ cum_tx + I(t/2) + baseline_exac,
                  id      = id,
                  corstr = "ar1",
                  data    = d)
```

Finally, we apply a model which takes into account both effects, applying AR(1) variance structure as well as MSM weights, together with a fully naïve model that uses neither.

```
# MSM with AR(1) correlation (handles both confounding and within-patient
# correlation)
msm_ar1_fit <- geeglm(exac ~ cum_tx + I(t/2) + baseline_exac,
                      id      = id,
                      weights = sw,
                      corstr = "ar1",
                      data    = d)

# Naive: no weights, independence correlation
naive_fit <- geeglm(exac ~ cum_tx + I(t/2) + baseline_exac,
                   id      = id,
                   corstr = "independence",
                   data    = d)

true_effect <- -2.0 # negative: treatment reduces exacerbations
est_msm     <- coef(msm_fit)["cum_tx"]
est_ar1     <- coef(ar1_fit)["cum_tx"]
est_ar1_msm <- coef(msm_ar1_fit)["cum_tx"]
est_naive   <- coef(naive_fit)["cum_tx"]

results <- data.frame(
  'True Effect'      = true_effect,
  'MSM with AR(1)'  = est_ar1_msm,
  'AR(1) Only'      = est_ar1,
  'MSM Only'        = est_msm,
  'Naive'            = est_naive,
  'check.names'     = FALSE
)

results
```

The resulting output follows:

```
results
      True.Effect MSM.with.AR.1. AR.1..Only MSM.Only   Naive
cum_tx          -2      -2.029492  -1.693832  -1.421701  -1.055598
```

Note that the joint MSM estimation with AR(1) autocorrelation results in the most accurate

treatment effect estimate. The other models all systematically underestimate the treatment effect. This occurs since a degree of the effect of treatment is ignored beyond the immediate subsequent timepoint: The models which do not include MSM weights ignore the causal path which connects A_1 to X_{m2} through A_2 , denoted using the dotted line in Figure 3.3. Conversely, the models which do not appropriately model the time lag lose the paths which connect the time-varying confounders to one another independently of treatment, such as those denoted using the dashed line in Figure 3.3.

3.5 G-estimation and Structural Causal Models

In some cases, even the stabilized weighting strategy might fail; G-estimation and Structural Causal Modelling have been developed for this case. To see where stabilized weights do not save us, we will continue with the asthma example, where a physician - let's call her Dr. Yersin - treats patients with severe asthma in an intensive care unit. The hospital protocol strongly recommends high-dose steroids whenever a patient's peak flow drops below 200 L/min, making treatment assignment nearly deterministic. When Marcus arrives with a peak flow of 180, Dr. Yersin prescribes high-dose steroids with probability 0.95 ($\pi(\text{high dose} | X = 180) = 0.95$). For patients with peak flow above 250, this probability drops to 0.05. Suppose there is also a second patient Livia, on whom we observed peak flow 185 - however, Livia happened to not receive steroids due to an insurance anomaly.

Suppose we attempt to use marginal structural models here. The stabilized weights become extreme; the weight we assign to Marcus' datapoint would be approximately $\omega = \frac{0.5}{0.95} \approx 0.53$. On the other hand, the stabilized weight for Livia's observation would be $\omega = \frac{0.5}{0.05} = 10$, meaning her single observation receives ten times the influence of a typical patient. With several such patients, our MSM estimates become dominated by a handful of observations with extreme weights, leading to high variance and unreliable treatment effect estimates.

Moreover, MSMs target a specific estimand that may not align with our scientific question. The MSM estimates what would happen if we randomly assigned treatment with probability 0.5 to all patients, regardless of their peak flow. But in practice, it is often desirable to estimate the effect of treatment for *specific patients* at different severity levels, either in lieu of or in addition to the marginal estimates. Would high-dose steroids help a patient with peak flow 250 as much as one with peak flow 180? The MSM cannot detect this, since it produces only an average estimate across all patients. MSMs can only tell us general facts about the drug, leading to conclusions about *general* optimal strategy - we need different methods to tell us about optimal strategies for individual patients. G-estimation has been developed in tandem with the structural nested model to this end (see [130, 131]).

The structural nested model [131] posits that Marcus's outcome can be decomposed as $r_j = r_j^{(0)} + \psi(a_j, x_j; \beta)$, where $r_j^{(0)}$ represents his counterfactual outcome had he not received steroids and $\psi(a_j, x_j; \beta)$ represents the causal effect of the steroids given his peak flow x_j . The function ψ is called the blip function, as it represents the "blip" or change in outcome attributable to treatment. The blip function can depend on patient-specific covariates such as peak flow, as in Equation 3.10.

$$\psi(a, x; \beta) = a \cdot (\beta_0 + \beta_1 \cdot \text{peakflow}) \quad (3.10)$$

To the Reinforcement Learning practitioner: The blip function is very similar to the Advantage score. The details of this overlap are discussed in Section 5.7.

In order for one to accurately estimate the parameter set β , the counterfactual untreated outcome $r_j^{(0)} = r_j - \psi(a_j, x_j; \beta)$ must be independent of treatment assignment conditional on covariates [172]; counterfactual potential outcomes are *ignorable* or Missing at Random. We can typically find unbiased estimates $\hat{\beta}$ of β by setting $E[h(x_j)(r_j - \psi(a_j, x_j; \hat{\beta}))(a_j - E[a_j | x_j])] = 0$ for some function h of the covariates, and then algebraically solving for $\hat{\beta}$. Notice that the treatment enters only through the centering term $a_j - E[a_j | x_j]$. For a binary treatment $E[a_j | x_j]$ is precisely the propensity score, so g-estimation does still require a model for treatment assignment. What it avoids is *inverting* that model. The propensity appears linearly in the estimating equation rather than in a denominator, so a near-deterministic treatment yields a centering term close to zero rather than a weight that explodes. G-estimation therefore sidesteps the extreme-weight pathology without sidestepping the need to model π .

The time-variant version of g-estimation is very similar. Suppose in Marcus’s case, after initial stabilization with steroids at visit 1, Dr. Yersin must decide on maintenance therapy at visits 2, 3, and beyond. For the sequence of treatments $\bar{a}_j = (a_0, a_1, \dots, a_j)$, the structural nested mean model [131] decomposes the outcome as $r_j = r_j^{(0)} + \sum_{k=0}^j \psi_k(a_k, \bar{x}_k; \beta)$, where each ψ_k captures the effect of treatment at time k on the final outcome. The g-estimation procedure solves for β by ensuring that at each time k , the partially demediated outcome is independent of treatment a_k given the history.

Judea Pearl [117] developed, largely in parallel with this line of work, a framework of structural *causal* models (SCMs) whose graphical language is broad enough to express both MSMs and g-estimation. In brief, a structural causal model consists of a set of structural equations. For our asthma example, the structural equations might be $\text{PeakFlow}_{j+1} = f_{\text{PF}}(\text{PeakFlow}_j, \text{Steroids}_j, \epsilon_j^{\text{PF}})$ and $R_j = f_R(\text{PeakFlow}_j, \epsilon_j^R)$. Causal Inference experts use the *g-formula* [130] to formalize the general question “What would happen to the population if everyone followed treatment regime $\bar{a}$?”; the g-formula is expressed in Equation 3.11. Note that the left-hand side is an *interventional* quantity: it is the mean outcome when the treatment regime is set by intervention, which is in general not the same as the conditional mean $E[R | \bar{A} = \bar{a}]$ obtained by passively observing subjects who happened to follow that regime. The Reinforcement Learning counterpart of the g-formula is the value function V^π , which likewise averages returns over the trajectory distribution induced by following a fixed policy. It is the blip function introduced above, not the g-formula, that corresponds to the *advantage* $A^\pi(x, a) = Q^\pi(x, a) - V^\pi(x)$; advantage learning, an agentic method of optimal control built on that quantity, is discussed in detail in Chapter 5.

$$E[R | \pi(\bar{A} = \bar{a})] = \sum_{\bar{x}} E[R | \bar{X} = \bar{x}, \bar{A} = \bar{a}] \prod_j P(X_j | \bar{X}_{j-1}, \bar{A}_{j-1} = \bar{a}_{j-1}) \quad (3.11)$$

In practice, whether to choose an MSM or g-estimation method often depends on the research question and data structure [39]. Generally speaking, if the treatment variation is substantial and you want only marginal effects for policy decisions, then the MSM suffices. If treatment is nearly deterministic, or if effect modification is of scientific interest, then one should use G-estimation.

3.6 Directed Acyclic Graphs

Causal diagrams such as Figures 2.2, 3.1, and 3.3 are critical components of a thorough causal analysis. Formally speaking, these causal diagrams are a type of *directed acyclic graph* (DAG).

Historical Note: Some of the earliest recorded DAGs appear in the work of Barbara Stoddard Burks (1902-1943), an American psychologist who made pioneering contributions to behavioral genetics and statistical techniques, particularly regarding the nature-nurture debate on intelligence. Burks would go on to mentor a young Claude Shannon for his summer internship at the Carnegie Institute for Washington in 1939; Shannon's subsequent thesis, "An Algebra for Theoretical Genetics" [154] is largely based on Burks' research about partial correlations [29, 30] though her name is markedly absent from the list of references, likely due to the sociopolitical implications of citing a woman researcher at the time. Later, Shannon's revolutionary work on information theory [153, 50] (with Burks' contributions also unattributed) would create rippling effects through many of its adjacent fields, including engineering, control theory, and physics; we will discuss in later chapters the contributions of Richard Bellman, who interacted with Shannon in the 50s and invented the foundational methods of Reinforcement Learning [14].

Burks herself suffered an unstable career, never achieving tenure despite her brilliance. In May 1943, she committed suicide by throwing herself from the George Washington Bridge [82].

A DAG consists of *nodes* (vertices) representing variables and *edges* (arrows) representing direct causal relationships. In our notation, we use capital letters for nodes (e.g., A , X , R) and arrows $\rightarrow$ to denote direct causal effects.

Consider the confounding diagram from Figure 3.1, reproduced conceptually here:

$$\begin{array}{l} X_{1m} \rightarrow A_2 \rightarrow R \\ X_{1m} \rightarrow R \end{array}$$

In this DAG, the unobserved confounder X_{1m} is a *parent* of both A_2 and R - we say that there is a *fork* in the DAG at the X_{1m} node. The second action, A_2 is called a *child* of X_{1m} because it is a downstream causal effect of X_{1m} . Similarly, we say that A_2 is a *parent* of R . Moreover, $X_{1m} \rightarrow A_2 \rightarrow R$ forms a *path* from X_{1m} to R ; this path is an *indirect* path because it is a composite of two edges. The path $X_{1m} \rightarrow R$ is a *direct* path.

A path is *directed* if all arrows point in the same direction along the path. The path $X_m \rightarrow A \rightarrow R$ is directed, while a path like $A \leftarrow X_m \rightarrow R$ is not. Note the difference between a *direct* path and a *directed* path.

A DAG encodes a network of statistical associations between variables. Two variables can be associated through chains, forks, or colliders. For example, a path $X_1 \rightarrow X_2 \rightarrow X_3$ would be a chain; here X_1 affects X_3 indirectly through X_2 . If X_2 caused both X_1 and X_3 : $X_1 \leftarrow X_2 \rightarrow X_3$ then the DAG would form a *fork*. The common cause X_2 creates association between X_1 and X_3 .

Finally, we have *colliders*, positioned as $X_1 \rightarrow X_2 \leftarrow X_3$. Unlike chains and forks, colliders do *not* create association between X_1 and X_3 unless we condition on X_2 (or its descendants). The collider structure deserves special attention because it behaves counterintuitively. In our smoking status confounding example, if we incorrectly controlled for

a variable that was caused by both age and drug efficacy (for example, exercise level), we would be “conditioning on a collider” and potentially introduce *spurious* associations between variables that are not causally related.

The most important application of DAGs in our context is determining when we can identify causal effects from observational data. The *backdoor criterion* provides sufficient conditions for estimating the causal effect of treatment A on outcome R .

Definition 3.9 (Backdoor Criterion) *A set of variables X satisfies the backdoor criterion relative to (A, R) if:*

1. *No variable in X is a descendant of A*
2. *X blocks every path from A to R that contains an arrow pointing into A*

Formally speaking, we say that a path between two variables is *blocked* if the following criteria are met:

- The path contains a chain $X \rightarrow C \rightarrow Y$ and we condition on C , or
- The path contains a fork $X \leftarrow C \rightarrow Y$ and we condition on C , or
- The path contains a collider $X \rightarrow C \leftarrow Y$ and we do *not* condition on C or any of its descendants

Paths that point into A are called *backdoor paths* because they represent non-causal associations between A and R . When X satisfies the backdoor criterion, we can estimate the causal effect by conditioning on X :

$$E[R \mid \pi(A = a)] = \sum_x E[R \mid A = a, X = x]P(X = x)$$

Let us return once more to the asthma example, with two more variables: X_{age} denoting the patient’s age and X_{sev} denoting the patient’s baseline asthma severity. Let X_{smoke} denote the patient’s smoking status. Suppose the true causal structure is as pictured in Figure 3.4. Here, the path $A \leftarrow X_{smoke} \rightarrow R$ is a fork, creating spurious association between treatment and outcome through the common cause X_{smoke} . The path $A \leftarrow X_{smoke} \leftarrow X_{age} \rightarrow X_{sev} \rightarrow R$ is a second backdoor path, containing a chain through X_{smoke} and a fork at X_{age} . Both of these structures complicate the estimation of the direct effect $A \rightarrow R$ that we wish to evaluate.

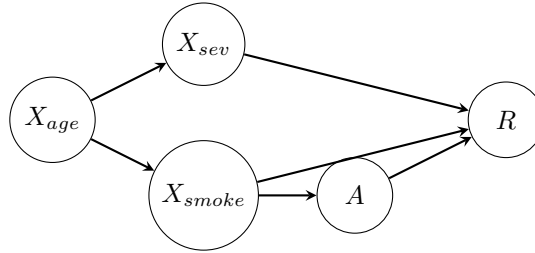


FIGURE 3.4: Directed Acyclic Graph with a chain, fork, and backdoor path.

To identify the causal effect of A on R , we must block both backdoor paths: $A \leftarrow X_{smoke} \rightarrow R$, and $A \leftarrow X_{smoke} \leftarrow X_{age} \rightarrow X_{sev} \rightarrow R$.

It is sufficient to condition on X_{smoke} to achieve valid treatment effects, since this would block the path $A \leftarrow X_{smoke} \rightarrow R$ as well as $A \leftarrow X_{smoke} \leftarrow X_{age} \rightarrow X_{sev} \rightarrow R$. We could

additionally condition on X_{age} , which may reduce the variance of the estimate but is not strictly necessary, since the A node is not on a path between X_{sev} and R . For the same reason, conditioning on X_{sev} alone would not remove the problem.

Now consider our Brazilian Jiu-Jitsu example. Suppose we wish to understand the causal effect of fatigue on submission success ($R_{success}$). It is simpler to work with positive associations for illustration, so we will consider the inverse of fatigue, energy (X_{energy}) instead. Suppose the causal structure follows the form depicted in Figure 3.5, where X_{skill} represents the player's skill and $X_{attempts}$ represents the number of submission attempts. In this scenario, players who are more skilled or less fatigued will attempt more submissions, but attempting a lot of submissions doesn't necessarily make you more likely to succeed at them.

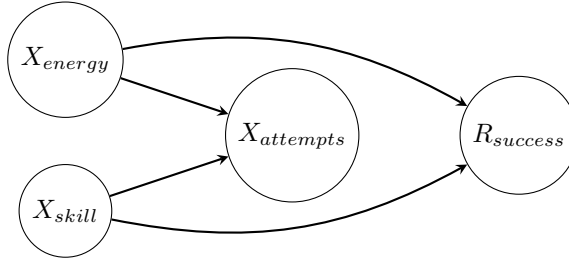


FIGURE 3.5: Directed Acyclic Graph for the BJJ collider example. Both X_{energy} and X_{skill} cause $X_{attempts}$, making it a collider. Note that $X_{attempts}$ has no outgoing edges.

Here, $X_{attempts}$ is a **collider**, since both X_{energy} and X_{skill} have arrows pointing into it. If we naively condition on $X_{attempts}$ (e.g., by only analyzing matches with similar numbers of attempts), we create a spurious *negative* association between energy and skill. To see this, consider a restricted population of players who attempted a lot of submissions. Since low skillset players who are fatigued are much less likely to be able to attempt submissions, most of the fatigued players in this group will be high-skill. No matter how tired a black belt is, s/he is still infinitely more likely to hit a submission than a white belt is; hence, among this restricted population, a spurious correlation between fatigue and success arises. In general, this spurious correlation is known as *collider bias*: we incorrectly estimate the effect of fatigue on success because, in conditioning on submission frequency, we've inadvertently created a positive association between fatigue and skill.

If we want to estimate the effect of fatigue, we should not condition on $X_{attempts}$, since conditioning on $X_{attempts}$ would unblock the collider path $X_{energy} \rightarrow X_{attempts} \leftarrow X_{skill} \rightarrow R_{success}$. One might condition on skill to reduce variance, but it is not required in order to obtain an accurate causal effect estimate.

The DAG framework extends naturally to longitudinal settings by creating time-indexed nodes as in Figure 3.3. The challenge in time-varying settings is that standard adjustment (conditioning on all confounders) fails when intermediate variables are both confounders and mediators. Consider the following DAG (Figure 3.6):

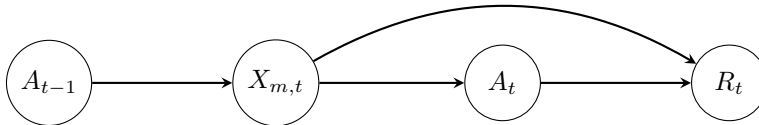


FIGURE 3.6: DAG with time-dependent confounding. $X_{m,t}$ is simultaneously a confounder for $A_t \rightarrow R_t$ and a mediator for $A_{t-1} \rightarrow R_t$.

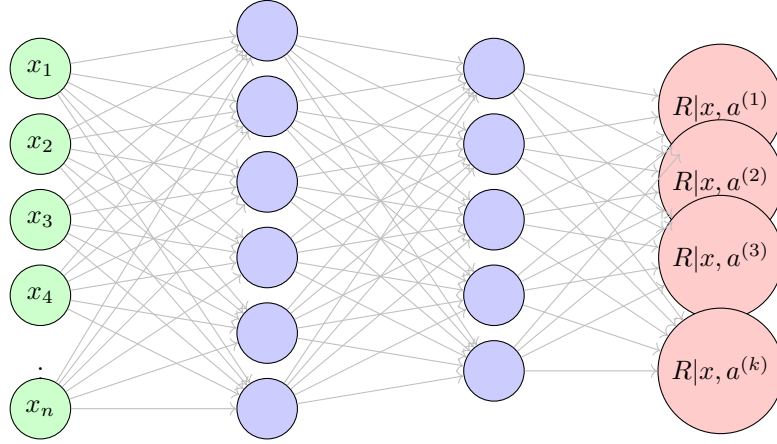


FIGURE 3.7: Deep Neural Network architecture. The network takes state features $x = (x_1, \dots, x_n)$ as input and outputs conditional estimates $R|x, a^{(k)}; \theta$ for each action $a \in \mathcal{A}$ given parameters θ used by the nodes.

Here $X_{m,t}$ is a confounder for the effect of A_t on R_t (we should control for it), but it is also a mediator for the effect of A_{t-1} (we should not control for it, as this would block part of the causal effect). Standard regression cannot handle this simultaneously. The MSM weights effectively create a pseudo-population where the backdoor paths through $X_{m,t}$ are blocked *without* conditioning on $X_{m,t}$ directly, thus preserving the mediated effects of prior treatments.

Requiring a causal diagram to be *acyclic* means that we assume causality only moves in one direction: Future events cannot affect the past. For instance, in physiological systems where blood pressure affects heart rate which in turn affects blood pressure, a static DAG is insufficient. However, we can approximate such systems by discretizing time and creating time-indexed DAGs where $X_t \rightarrow X_{t+1}$, effectively “unrolling” the feedback loop across time; we will discuss this in more detail in Section 6.4 on non-linear modelling.

Modeling and analysis of DAGs increases exponentially in difficulty as the size and complexity of the system increases. How, then, can we model a system such as a complex biological process with hundreds of nodes? Fortunately, computer scientists have developed exactly such a method: By leveraging the *acyclic* nature of the causal diagram, one can use an iterative estimation method called *backpropagation* to estimate a model such as the one depicted in Figure 3.7. We will return to these later in our discussion of a popular Reinforcement Learning technique called *Deep Q-learning*.

3.7 Instrumental Variables

IPTW, MSMs, and g-estimation all require that we observe and correctly specify the confounders blocking backdoor paths. But what if such a blockage is impossible? In many observational settings, the confounders are unmeasured, unmeasurable, or so numerous that any adjustment set is implausible. One compelling example is in genetics, where one might encounter 18,000+ genes in a dataset. The analysis of genetic causal effects is particularly complex because genes do not function independently, rather they group together in

clusters of inter-related gene modules to cause different phenotypes and responses. Instrumental variable (IV) methods offer an alternative identification strategy to modeling; this alternative can be useful for isolating specific effects in complex datasets such as the genetic example. The idea is to exploit an external source of variation that shifts treatment without contamination from the confounding variable.

An instrumental variable x_{iv} allows us to identify the causal effect of treatment A on outcome R when A shares unmeasured common causes with R . If we can find a variable that affects R *only through* its effect on A , then the variation in R induced by x_{iv} must be causally attributable to A .

Definition 3.10 (Instrumental Variable) *A variable x_{iv} is a valid instrument for the causal effect of A on R iff all three of the following conditions hold.*

1. **Relevance:** x_{iv} is associated with A ; i.e., $x_{iv} \not\perp A$
2. **Exclusion Restriction:** x_{iv} affects R only through A ; there is no direct path $x_{iv} \rightarrow R$
3. **Independence:** x_{iv} is independent of all unmeasured confounders of the $A \rightarrow R$ relationship

The DAG representation clarifies these conditions. Figure 3.8 shows the instrumental variable structure, where x_m represents unmeasured (missing) confounders.

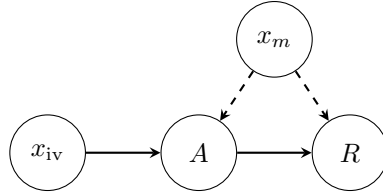


FIGURE 3.8: Directed Acyclic Graph illustrating an instrumental variable. The instrument x_{iv} affects treatment A but has no direct path to outcome R . The unmeasured confounder x_m creates a backdoor path that cannot be blocked by conditioning.

Note that the backdoor path $A \leftarrow x_m \rightarrow R$ in Figure 3.8 cannot be blocked by conditioning, since x_m is unmeasured. However, the instrument x_{iv} can save us, since x_{iv} shifts A without traveling through x_m . The x_{iv} -induced variation in A is therefore “clean” (uncontaminated by the confounder), and so manipulating it can give us the complete picture of the causal effect in question.

A classic application of instrumental variables in biomedical research is *Mendelian randomization* (MR), which uses genetic variants as instruments for modifiable exposures. Here, alleles are randomly assigned at meiosis (cell division), independently of the environmental and behavioral confounders that frequently occur in observational studies. Hence, if a genetic variant x_{iv} affects an exposure A (e.g., LDL cholesterol levels) and has no effect on the outcome R (e.g., coronary heart disease) except through A , then x_{iv} serves as a valid instrument.

Suppose we wish to test whether elevated LDL cholesterol causes coronary heart disease, or whether the observed association is confounded by diet, exercise, socioeconomic status, and other lifestyle factors. Standard regression adjusting for measured confounders may be biased if important confounders are unmeasured.

Let x_{iv} represent a genetic variant in the *PCSK9* gene known to affect LDL cholesterol levels. We can verify that each IV condition holds as follows.

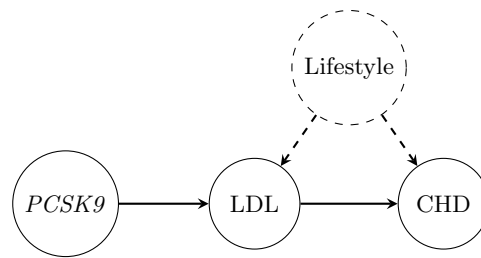


FIGURE 3.9: Mendelian randomization for the LDL-CHD relationship. The *PCSK9* genetic variant serves as an instrument: it affects LDL cholesterol levels but has no direct effect on coronary heart disease. Lifestyle factors (diet, exercise, socioeconomic status) confound the LDL-CHD relationship but are independent of genotype, which is assigned at conception.

1. **Relevance:** Individuals carrying certain *PCSK9* variants have measurably lower LDL cholesterol.
2. **Exclusion:** The genetic variant affects heart disease risk only through its effect on LDL, not through other pathways (this is the key assumption requiring domain knowledge).
3. **Independence:** Genetic variants are assigned at conception, prior to and independently of lifestyle confounders.

Therefore the *PCSK9* gene variant is a valid instrumental variable (Figure 3.9). An R simulation of the LDL-PCSK9 dataset follows.

Simulating Mendelian Randomization Data

```

# mendelian_randomization.R
set.seed(2024)
n <- 5000

# Unmeasured confounder (lifestyle factors)
xm <- rnorm(n, 0, 1)

# Genetic instrument: binary SNP (0 = reference, 1 = variant)
# Assigned independently of xm
xiv <- rbinom(n, 1, 0.3)

# LDL cholesterol: affected by both genetics and lifestyle
# xiv reduces LDL; xm (poor lifestyle) increases LDL
LDL <- 130 + (-15) * xiv + 20 * xm + rnorm(n, 0, 10)

# Coronary heart disease risk score
# True causal effect of LDL: 0.5 per unit
# xm also directly affects CHD (confounding)
CHD <- 10 + 0.5 * LDL + 15 * xm + rnorm(n, 0, 5)

data <- data.frame(xiv = xiv, LDL = LDL, CHD = CHD, xm = xm)
  
```

The naive regression of CHD on LDL yields a biased estimate because it conflates the causal effect with confounding through x_m :

Naïve OLS Estimate (Confounded)

```
# Naive estimate: regress CHD on LDL
naive_model <- lm(CHD ~ LDL, data = data)
coef(naive_model)["LDL"]
```

```
> coef(naive_model)["LDL"]
      LDL
1.048168
```

The naive estimate of approximately 1.048 substantially overstates the true causal effect of 0.5, because patients with high LDL also tend to have poor lifestyle factors (high x_m) that independently increase CHD risk.

Two-stage least squares (2SLS) provides a consistent estimate by using only the x_{iv} -induced variation in LDL; we can implement it as follows.

Two-Stage Least Squares Estimation

```
# mendelian_randomization.R
library(ivreg)

# 2SLS: instrument LDL with xiv
iv_model <- ivreg(CHD ~ LDL | xiv, data = data)
summary(iv_model)
```

```
> coef(iv_model)["LDL"]
      LDL
0.5150925
```

The IV estimate of approximately 0.515 recovers the true causal effect. The `ivreg` package (or equivalently, `AER::ivreg`) implements 2SLS by first regressing the endogenous variable on the instrument(s), then regressing the outcome on the fitted values from the first stage. Carrying those two stages out by hand makes the mechanism explicit, and gives us the fitted first-stage object we will need for the diagnostics below.

Two-Stage Least Squares, By Hand

```
# mendelian_randomization.R

# Stage 1: regress the endogenous variable on the instrument
stage1 <- lm(LDL ~ xiv, data = data)
data$LDL_hat <- fitted(stage1)

# Stage 2: regress the outcome on the fitted values
stage2 <- lm(CHD ~ LDL_hat, data = data)
coef(stage2)["LDL_hat"]
```

```
> coef(stage2)["LDL_hat"]
      LDL_hat
0.5150925
```

The hand-rolled estimate agrees with `ivreg` to every printed digit. Note that while this manual approach yields the correct point estimate, the standard errors from `stage2` are incorrect because they fail to account for the uncertainty in the first-stage prediction. The practitioner should therefore use dedicated IV software for inference whenever possible.

The perceptive reader will notice that under treatment effect heterogeneity, the estimand recovered is not the average treatment effect (ATE) but rather a *local* average treatment effect (LATE), i.e. the effect among a specific subpopulation whose treatment status is shifted by the instrument. To understand this subtlety, consider four different classes of subjects, with $A(x_{iv})$ denoting the treatment that subject j would receive if the instrument were set to x_{iv} (Definition 3.11).

Definition 3.11 (Compliance Types (Angrist, Imbens & Rubin, 1996 [2])) For a binary instrument x_{iv} and binary treatment A :

- **Compliers:** $A(1) = 1$ and $A(0) = 0$. Treatment status follows the instrument.
- **Always-takers:** $A(1) = 1$ and $A(0) = 1$. Always treated regardless of x_{iv} .
- **Never-takers:** $A(1) = 0$ and $A(0) = 0$. Never treated regardless of x_{iv} .
- **Defiers:** $A(1) = 0$ and $A(0) = 1$. Treatment status opposes the instrument.

See also Figure 3.10.

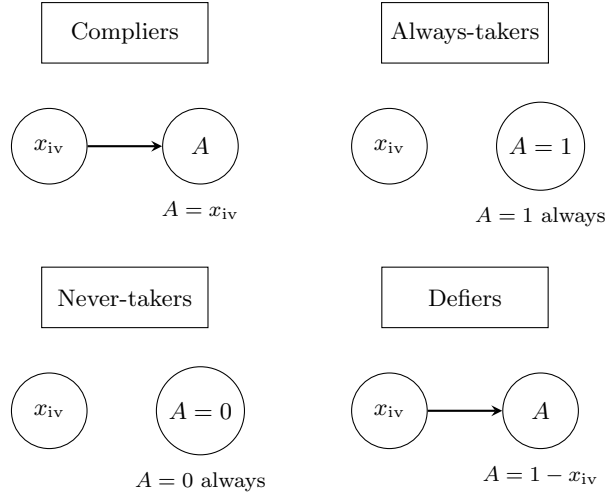


FIGURE 3.10: The four compliance types under a binary instrument x_{iv} and binary treatment A . Only compliers contribute to the LATE: they are the subpopulation whose treatment status is shifted by the instrument. The monotonicity assumption rules out defiers. Always-takers and never-takers receive the same treatment regardless of x_{iv} , so the instrument provides no information about their treatment effects.

Under the assumption that there are no defiers (also referred to as *monotonicity*), the IV estimator identifies the average treatment effect among compliers (Equation 3.12).

$$\widehat{\text{LATE}} = \frac{E[R|x_{iv} = 1] - E[R|x_{iv} = 0]}{E[A|x_{iv} = 1] - E[A|x_{iv} = 0]} = E[R(1) - R(0)|\text{Complier}] \quad (3.12)$$

Here, the denominator $E[A|x_{iv} = 1] - E[A|x_{iv} = 0]$ is the first-stage effect, i.e. the proportion of the population whose treatment is shifted by the instrument. This quantity appears in the denominator because the IV estimator “scales up” the reduced-form effect (the numerator) by the strength of the first stage. In our Mendelian randomization example, “compliers” are individuals whose LDL levels are actually affected by the genetic variant. The LATE represents the causal effect of LDL on CHD specifically for this subpopulation. If the treatment effect is homogeneous (constant across individuals), the LATE equals the ATE. But if effects vary (perhaps, for example, LDL reduction is more cardioprotective for some individuals than others) then the LATE may differ from the population average effect.

To the Reinforcement Learning Practitioner: The “local” nature of IV estimates is reminiscent of off-policy evaluation, where one can only learn about the effects of actions that were actually taken under some policy. IV identifies effects only for compliers (those whose treatment was shifted by the instrument) the same way that off-policy methods can only evaluate policies that have support under the behavior policy. Importance sampling weights in off-policy RL serve an analogous role to the first-stage denominator in IV, rescaling observed outcomes to account for the mismatch between the policy that generated the data and the policy being evaluated.

Whether the LATE is the estimand of scientific interest or not depends on one’s application. In Mendelian randomization, if the goal is to inform pharmaceutical development (e.g., evaluating drugs that mimic the effect of the genetic variant), the complier subpopulation may itself actually be the target population of interest. In other applications, extrapolating from the LATE to the ATE requires additional assumptions about treatment effect homogeneity.

The validity of IV estimation rests on assumptions that are largely untestable. However, one may perform a suite of diagnostic examinations in order to flag potential problems.

First, consider the required relevance condition, which demands that x_{iv} actually shifts A . When the first-stage relationship is weak, the IV estimator becomes severely biased toward the OLS estimate and has inflated variance. The standard diagnostic is the first-stage F-statistic testing the null that the instrument has no effect on the endogenous variable.

Testing for Weak Instruments

```
# mendelian_randomization.R

# F-statistic for instrument strength
summary(stage1)$fstatistic
```

```
> summary(stage1)$fstatistic
      value      numdf      dendif
470.4189    1.0000 4998.0000
```

The conventional rule (per [159]) is that an F-statistic below 10 indicates a weak instrument problem. Our simulated example has $F \approx 470$, indicating a strong first stage. In practice, Mendelian randomization studies often face weak instrument concerns when the genetic variant explains only a small fraction of variance in the exposure.

If instruments are weak, then the researcher has a few different options available, though the case of weak instruments is still an open area of research. One might, for example, apply weak-instrument-robust inference methods such as the Anderson-Rubin test or the conditional likelihood ratio test; an alternative is to report the limited information maximum

likelihood (LIML) estimator, which has better finite-sample properties than 2SLS under weak instruments. See also [161] for details and additional advice on detecting weak instruments in practice.

Anderson-Rubin Confidence Interval

```
# Weak-instrument-robust inference
library(ivmodel)

iv_obj <- ivmodel(Y = data$CHD, D = data$LDL, Z = data$xiv)
AR.test(iv_obj)
```

The Anderson-Rubin test inverts the reduced-form regression to construct confidence intervals that remain valid even when instruments are weak. The cost is wider intervals compared to standard 2SLS inference when instruments are strong.

The exclusion restriction, which requires that x_{iv} affects R only through A , cannot be tested with a single instrument. However, when multiple instruments are available, *overidentification tests* can detect violations. The idea is that if two instruments $x_{iv,1}$ and $x_{iv,2}$ are both valid, they should yield the same causal estimate, at least up to sampling variability (see Figure 3.11). If they disagree substantially, then at least one must violate the exclusion restriction.

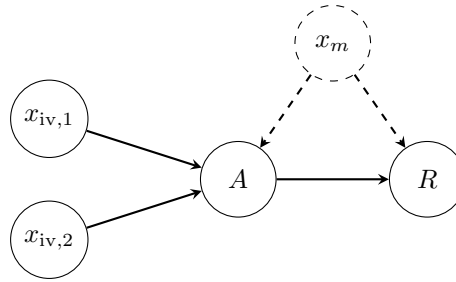


FIGURE 3.11: Multiple instruments for the same endogenous variable. When two or more instruments $x_{iv,1}$ and $x_{iv,2}$ are available, the model is overidentified: each instrument implies a separate estimate of the causal effect. The Sargan-Hansen test exploits this redundancy to test whether all instruments satisfy the exclusion restriction. If instruments disagree, at least one must be invalid.

The Sargan-Hansen J-statistic is another useful device; this statistic allows us to test whether the overidentifying restrictions are satisfied. A significant p-value suggests that at least one instrument is invalid. However, failure to reject does not confirm validity, since all instruments could be invalid in the same direction.

Sargan-Hansen Overidentification Test

```
# Simulate a second instrument
data$xiv2 <- rbinom(n, 1, 0.4)
data$LDL <- 130 + (-15) * data$xiv + (-10) * data$xiv2 +
            20 * data$xv + rnorm(n, 0, 10)
data$CHD <- 10 + 0.5 * data$LDL + 15 * data$xv + rnorm(n, 0, 5)

# 2SLS with multiple instruments
iv_overid <- ivreg(CHD ~ LDL | xiv + xiv2, data = data)
summary(iv_overid, diagnostics = TRUE)
```

Note that the validity of the LATE interpretation still requires the data to be *monotonic*: Defiers must not exist. In medical applications, this assumption is often plausible, since it would be unusual for a genetic variant that generally lowers LDL to paradoxically raise it in some individuals. However, in social science applications with behavioral instruments, defiance can be a grave concern. When monotonicity is suspect, bounds approaches can partially identify treatment effects without the assumption, though the resulting intervals are typically wide. If monotonicity could be plausibly violated, then methods other than instrumental variables should be considered.

Table 3.7 summarizes how instrumental variables relate to the other identification strategies discussed in this chapter.

TABLE 3.7: Comparison of identification strategies for causal effects

Method	Key Assumption	When to Use	Estimand
Covariate Adjustment	All confounders observed and correctly specified	Confounders measurable; backdoor criterion satisfied	ATE or CATE
IPTW/MSM	Positivity; correct propensity model	Time-varying confounding; marginal effects desired	ATE
G-estimation	Correct blip function specification	Near-deterministic treatment; effect modification of interest	CATE
Instrumental Variables	Valid instrument exists; exclusion restriction holds	Unmeasured confounding; natural experiment available	LATE

The choice between methods depends on which assumptions are most credible in a given application. IV methods trade the assumption that all confounders are observed (required for backdoor adjustment) for the assumption that a valid instrument exists (exclusion restriction). Neither assumption is directly testable; the analyst must rely on domain knowledge and auxiliary evidence.

When both approaches are feasible, applying multiple methods and checking for concordance provides valuable robustness. Substantial disagreement between IV and covariate-adjusted estimates suggests that the estimation is affected by either i) unmeasured confounding (biasing the adjusted estimate) or ii) instrument invalidity (biasing the IV estimate). It is up to the researcher to determine whether i) or ii) (or both) is most likely to be the

problem.

In practice, Mendelian randomization studies rarely rely on a single genetic variant; our example is simplified for pedagogical illustration. More realistically, Genome-wide association studies have identified numerous variants associated with most exposures of interest, raising the question of how to combine information across instruments. Using multiple instruments improves precision by exploiting more identifying variation, however, the primary concern in MR is a phenomenon called *pleiotropy*. Pleiotropy is the possibility that genetic variants affect the outcome through pathways other than the exposure, violating the exclusion restriction. Methods such as MR-Egger regression [22], weighted median estimators [23], and mode-based estimation [56] have been developed to provide valid inference under various assumptions about the proportion and nature of invalid instruments; interested readers may find a more detailed overview in [145].

The IV framework also extends to time-varying treatments, though the notation and assumptions become more complex. In a two-period setting with treatment at times 1 and 2, we may have instruments $x_{iv,1}$ and $x_{iv,2}$ for each treatment decision. The structural nested mean models discussed in Section 3.5 can incorporate instrumental variables through *g-estimation with instruments*, which combines the blip function approach with IV-style identification. A full treatment of longitudinal IV methods is beyond our scope here; interested readers should consult [61] for the epidemiological perspective and Chapter 11 of [184] for the econometric treatment.

3.8 Further Reading

Rosenbaum & Rubin’s work “The Central Role of the Propensity Score in Observational Studies for Causal Effects” [134] introduced the propensity score methodology that underpins much of modern observational Causal Inference. Their framework provides the theoretical foundation for IPTW and related methods discussed in this chapter.

The book *Causality: Models, Reasoning, and Inference* by Judea Pearl [117] offers one of the most comprehensive treatments of causal diagrams and the graphical approach to identifying confounding relationships. Chapter 3 on “Causal Diagrams and the Identification of Causal Effects” is particularly relevant to understanding when IPTW methods will succeed or fail.

For readers seeking a more statistical perspective on confounding, the aforementioned text *Causal Inference: What If* provides an exceptional modern treatment [62]. Their discussion of exchangeability, positivity, and consistency assumptions (Chapters 1-3) directly connects to the concepts presented here, while Chapters 12-13 cover IP weighting with worked examples.

The original EM algorithm was introduced by Dempster, Laird, and Rubin [41], who showed that it converges to local maxima of the likelihood function under general conditions. For missing data applications specifically, Little and Rubin [98] provide comprehensive coverage of EM-based approaches. In the context of Causal Inference, Seaman and White [151] review how multiple imputation methods like EM can be used to handle missing confounders, while Mayer et al. [104] demonstrate how EM-based imputation can be combined with propensity score methods for doubly robust Causal Inference.

[8] provides a practical, accessible introduction to propensity score methods beyond basic IPTW, including matching, stratification, and doubly robust approaches. The variance properties of IPTW estimators, particularly the extreme weight problem highlighted in the BJJ example, are thoroughly analyzed in [80], which also introduces doubly robust

methods that combine propensity scores with outcome modeling. [102] provides theoretical foundations for stabilized weights and compares various propensity score approaches, with particular attention to finite sample properties.

“Marginal Structural Models and Causal Inference in Epidemiology” [133] is the foundational paper introducing MSMs. “Marginal Structural Models to Estimate the Causal Effect of Zidovudine on the Survival of HIV-Positive Men” [63] demonstrates the practical application of MSMs to HIV research, showing how standard methods can severely underestimate treatment effects when time-varying confounders are present.

A classic application of MSMs comes from HIV research [63]. Using observational data from the Multicenter AIDS Cohort Study, Hernán et al. evaluated the causal effect of zidovudine therapy on mortality while accounting for time-dependent confounding by CD4 count. Their analysis revealed that the standard statistical methods underestimated the beneficial effect of zidovudine by failing to properly account for the time-varying nature of both the treatment and the confounders.

More recently, MSMs have been applied to evaluate dynamic treatment regimes in cancer, cardiovascular disease, and mental health, where treatment decisions constantly evolve in response to a patient’s changing condition.

Many excellent worked examples of classical Causal Inference, including calculation of average treatment effects, use of inverse probability weighting, and handling of confounding, are already available in the literature. Our asthma rate reduction example is based on the CD4 example found in Chapter 2 of “Causal Inference for Statistics, Social, and Biomedical Sciences” by Imbens and Rubin [69], which walks through a classical example using potential outcomes and randomized assignment. A translator between their notation and the unified notation in this book is available in Table 3.8. For details on the implementation of IPTW, see [37].

Concept	This Book’s Notation	Imbens & Rubin Notation	Comment
Outcome (random variable)	R	Y	“R” for “reward” (aligns with RL)
Treatment assignment (action)	A	T	“A” for “action” more general than “T” for “treatment”
Potential outcome under action $a^{(1)}$	$R \mid A = a^{(1)}$	$Y(0)$	$a^{(1)}$ is the reference arm
Potential outcome under action $a^{(2)}$	$R \mid A = a^{(2)}$	$Y(1)$	$a^{(2)}$ is the comparison arm
Observed outcome	R	Y_o	Same object
Counterfactual outcome	R_{cf}	$Y(1 - T)$ (implicit)	Made explicit here
Treatment policy	$\pi(a)$	$P(T = 1)$	Policy idea from RL
Average Treatment Effect (ATE)	$E[R \mid A = a^{(2)}] - E[R \mid A = a^{(1)}]$	$E[Y(1) - Y(0)]$	Same meaning
Ignorability assumption	$A \perp \{R a\} \mid X$	$T \perp \{Y(1), Y(0)\} \mid X$	Standard causal assumption

TABLE 3.8: Translation between this book’s notation and the standard notation of Imbens & Rubin (2015).

Statistical Analysis with Missing Data by Little and Rubin provides essential background on missing data mechanisms [98]. “Graphical Models for Processing Missing Data” [109] provides a graph-based treatment of the conditions under which imputation-based approaches to confounding will succeed.